

Exponential Equations and an Introduction to Logarithms (Homework)

Solve the following equations. Write the Roman numeral of each exponent rule that you use.

- ① $\sqrt[4]{7} = 7^x$
- ② $\frac{1}{27} = 3^x$
- ③ $8 = 8^{5-4x}$
- ④ $10^{3x-9} = 100^x$
- ⑤ $8^{-5x+2} = 2^{1-x}$
- ⑥ $1 = 3^{\frac{x}{2}+3}$
- ⑦ $5^{2x} = 25 \cdot 5^{x-4}$
- ⑧ $\frac{9^x}{9^{6x+2}} = 9^{8x}$
- ⑨ $216 = 2^x \cdot 3^x$
- ⑩ $81 = \frac{63^x}{7^x}$
- ⑪ $10^x = (\sqrt[5]{10})^2$
- ⑫ $5 = 125^{8x-8}$
- ⑬ $3^{2x-4} \cdot 3^{x-3} = 1$

$$\textcircled{14} \sqrt[5]{6} = \frac{6^x}{6^{3-2x}}$$

$$\textcircled{15} 5^x \cdot 3^x = \frac{1}{225}$$

$$\textcircled{16} \frac{72^x}{8^x} = \sqrt[3]{9^4}$$

$$\textcircled{17} \frac{1}{2} = 32^x$$

$$\textcircled{18} \frac{110^x}{10^x} = \frac{121}{11}$$

$$\textcircled{19} 1 = \sqrt[6]{4^x} \cdot 4^{-2x+5}$$

$$\textcircled{20} 35 = 7^{\frac{x}{3}} \cdot (\sqrt[3]{5})^x$$

Determine if you can solve the following equations without a calculator. If you need a calculator, write the solution in $\log_b C$ notation, and then use a calculator to approximate the answer to the nearest ten thousandth. Write the Roman numeral of each exponent rule that you use.

$$\textcircled{21} 35 - 6(4^x) = 23$$

$$\textcircled{22} -11 = -\frac{6^x}{2} + 7$$

$$\textcircled{23} 46 = 2(9^x) + 32$$

$$\textcircled{24} -5(2^x) + 1 = -14$$

$$\textcircled{25} 6 = 4(3^x) + 2$$

$$(26) -10 + \frac{7^x}{3} = -7$$

$$(27) 2(5^x) - 1 = 15$$

Answers

$$\text{IX } (1) \boxed{x = \frac{1}{4}}$$

$$\text{VIII } (2) \boxed{x = -3}$$

$$\text{VI } (3) \boxed{x = 1}$$

$$\text{V } (4) \boxed{x = 9}$$

$$\text{V } (5) \boxed{x = \frac{5}{14}}$$

$$\text{VII } (6) \boxed{x = -6}$$

$$\text{I } (7) \boxed{x = -2}$$

$$\text{II } (8) \boxed{x = -\frac{2}{13}}$$

$$\text{III } (9) \boxed{x = 3}$$

$$\text{IV } (10) \boxed{x = 2}$$

$$\text{X } (11) \boxed{x = \frac{2}{5}}$$

$$\text{V, VI } (12) \boxed{x = \frac{25}{24}}$$

$$\text{I, VII } (13) \boxed{x = \frac{7}{3}}$$

$$\text{IX, II } (14) \boxed{x = \frac{16}{15}}$$

$$\text{III, VIII } (15) \boxed{x = -2}$$

$$\text{IV, X' } (16) \boxed{x = \frac{4}{3}}$$

$$\text{VI, VIII, V } (17) \boxed{x = -\frac{1}{5}}$$

$$\text{IV, IX, II } (18) \boxed{x = \frac{3}{2}}$$

$$\text{I, VII, X' } (19) \boxed{x = \frac{30}{11}}$$

$$\text{III, VI, X } (20) \boxed{x = 3}$$

$$\text{V, VI or IX } (21) \boxed{x = \frac{1}{2}}$$

$$(22) \boxed{x = 2}$$

$$(23) \boxed{x = \log_9 7}$$

$$x \approx 0.8856$$

$$(24) \boxed{x = \log_2 3}$$

$$x \approx 1.585$$

$$\text{VII } (25) \boxed{x = 0}$$

$$(26) \boxed{x = \log_7 9}$$

$$x \approx 1.1292$$

$$(27) \boxed{x = \log_5 8}$$

$$x \approx 1.292$$